Computer Engineering Department

National Institute of Technology, Kurukshetra-136119

Mid term 1 , Even Semester 2024-25

Paper: Image Processing (CSPE 306)

Max Marks: 15 Total time: 50 m

1. (i) What are the main components of a general-purpose digital image processing system? Explain the relationship between the electromagnetic spectrum and image generation.

   (ii) Given a grayscale image on paper with a physical dimension of 4 inches × 3 inches, scanned at the rate of 200 dpi, calculate the following:

   (a) How many bits are required to represent the image?
   (b) How much time is required to transmit the image if the modem speed is 56 kbps? [5 marks]

2. A triangle is marked by the points (1,1), (5,5), and (10,10).

   (a) Apply a translation of $\Delta x = 3$ and $\Delta y = 4$. Explain the change.

   (b) Apply a rotation with angles 45°, 90°, 180°, and 270°. Explain the change. [3 marks]

3. Consider the following adjacent regions:

1 1 0 0	|	0 1 0 0
0 0 0 0	|	1 0 0 0
1 1 1 1	|	0 1 1 1
1 1 0 0	|	0 1 0 0
R_1		R_2

   What is the connectivity between these two regions? [4 marks]

4. Apply DFT to the following sequences:
   (a) {1, 2, 8, 9}

   Prove that the inverse transform works. What is the need for Fourier transforms? [3 marks]